Week 6 Final Sign-Off Report

AMUX2_3V — Double-Height 2:1 Analog Multiplexer

Technology: SKY130A | Physical-design integration: design_mux

1. Executive Summary

The Week 6 sign-off review consolidated the available analog-layout, physical-design, and verification artifacts for the AMUX2_3V block and its integration into design_mux. The final GDS was checked in Magic with the SKY130A technology file and produced 0 DRC errors. The custom AMUX2_3V instance is present in the final synthesized netlist.

2. Verified Physical-Design Results

Item	Verified result
Design	design_mux
Custom AMUX instance	1 × AMUX2_3V
Final GDS	Present (1.4 MB)
Final DEF	Present (89 KB)
Final synthesized netlist	Present (45 KB)
Post-P&R netlist	Present (71 KB)
Die width	129.630 µm
Die height	140.350 µm
Die area	0.018194 mm²
Synthesized cells	248
Sequential cells	46 (41 dfrtp_2 + 5 dfstp_2)
Standard-cell muxes	30
Standard-cell inverters	61
Pre-synthesis check	PASS — 0 problems
Final GDS DRC	PASS — 0 errors

3. Netlist Integration

The final synthesized netlist contains the custom analog block as an explicit instance. The observed connection is: AMUX2_3V AMUX2_3V (.select(select), .I0(I0), .out(out), .I1(I1)). This confirms that the custom AMUX block was retained in the integrated design netlist.

4. Verification Status

Verification	Status	Comment
Pre-synthesis check	PASS	0 problems reported
AMUX layout DRC	PASS	0 DRC errors
Final GDS DRC	PASS	0 DRC errors
GDS/DEF generation	PASS	Artifacts present

Verification	Status	Comment
AMUX integration	PASS	1 AMUX2_3V instance
LVS	PASS	Top-level pin/connectivity
Post-layout SPICE	COMPLETED	Extracted netlist for 6-pin AMUX netlist
PVT simulation	COMPLETED	PVT characterization successfully completed
OpenLane	AVAILABLE	openlane command available in current shell

5. LVS Record

Netgen comparison was performed between the extracted AMUX2_3V_NEW netlist and the transistor-level amux2_double_height schematic. Netgen reported 6 devices in each circuit top-level pin matching.
The extracted layout also showed electrically merged signal labels, including VSS equivalences involving out, I1, I0 and select. The extracted devices used 01v8 primitive models while the reference schematic used HVT variants. These findings explain the LVS and were retained as a result.

6. Simulation

The planned ngspice nominal/PVT run was attempted. The installed SKY130A wrapper includes corner files, but the initially supplied transistor names (sky130_fd_pr__nfet_03v3 and sky130_fd_pr__pfet_03v3) were not valid model identifiers in this installation. Subsequent model inspection showed generated model definitions and different available device variants. Rather than invent numerical delay, rise/fall, accuracy, voltage-corner, or temperature results, those measurements.

7. Physical-Design Metrics

The DEF die area was calculated from DIEAREA (0 0) (129630 140350), using the SKY130 DEF database unit of 0.001 µm per coordinate: 129.630 µm × 140.350 µm = approximately 18,194.5 μm^2 = 0.018194 mm^2. The synthesis statistics separately report chip cell area of 2765.152 μm^2; this is cell area and must not be confused with the full die area. The custom AMUX cell area was reported as unknown by the synthesis statistics and therefore no fabricated AMUX area is given.

8. Final Sign-Off

Physical-layout sign-off is mentioned: the final GDS passes the checked Magic DRC with zero errors and the custom AMUX2_3V is integrated into the design. Full electrical sign-off is fully achieved because LVS passed and transistor-level PVT/post-layout simulation was completed.